

Advanced (Habanero)

- Q1.** Which lines are parallel: $y = 2x + 1$ and $y = 2x - 3$?
(A) Yes
(B) No
(C) Maybe
(D) Insufficient info
(E) None
- Q2.** If $y = 3x - 2$ and $y = 3x + 1$, which is true?
(A) Perpendicular
(B) Parallel
(C) Skew
(D) Same line
(E) None
- Q3.** The equation of a line parallel to $y = x - 4$ is $y =$
(A) $x + 2$
(B) $x - 2$
(C) $2x - 1$
(D) $-x + 1$
(E) $x + 4$
- Q4.** Which of the following lines has the same slope as $y = -2x + 1$?
(A) $y = 2x + 1$
(B) $y = -2x - 3$
(C) $y = x - 2$
(D) $y = -x + 4$
(E) $y = x + 3$
- Q5.** The lines $y = x + 2$ and $y = -x + 1$ are
(A) Parallel
(B) Perpendicular
(C) Skew
(D) Same line
(E) None
- Q6.** If two lines have slopes m_1 and m_2 , when are they parallel?
(A) $m_1 = m_2$
(B) $m_1 \neq m_2$
(C) $m_1 = -m_2$
(D) $m_1 \cdot m_2 = 1$
(E) $m_1 + m_2 = 0$
- Q7.** The equation $y = 4x - 3$ has a slope of
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5
- Q8.** What is the slope of the line parallel to $y = -3x + 2$?
(A) -3
(B) 3
(C) 2
(D) 1
(E) 0

Open Ended Questions (FRQ)

- Q9.** Write the equation of a line parallel to $y = 2x - 1$ that passes through the point $(1, 3)$.
- Q10.** Given two parallel lines $y = x + 1$ and $y = x - 2$, find the distance between them.

Chef's Tips

- Tip 1: Use the slope formula to identify parallel lines
- Tip 2: To write equations of parallel lines, keep the same slope
- Tip 3: Analyze relationships between parallel lines using their slopes and y-intercepts